

03 — Intermediate SQL

This module builds on SQL Basics with powerful query patterns used daily by data engineers, backend developers, and data analysts.

Contents

File	Topic	Difficulty
01_joins_complete.md	INNER, LEFT, RIGHT, FULL, CROSS, SELF JOINS	Intermediate
02_aggregations.md	GROUP BY, HAVING, COUNT, SUM, AVG, MIN, MAX	Intermediate
03_subqueries.md	Scalar, row, table, correlated subqueries	Intermediate
04_exists_any_all.md	EXISTS, NOT EXISTS, ANY, ALL operators	Intermediate
05_set_operations.md	UNION, UNION ALL, INTERSECT, EXCEPT	Intermediate
06_case_expressions.md	CASE, COALESCE, NULLIF, conditional logic	Intermediate
07_grouping_sets_rollup_cube.md	GROUPING SETS, ROLLUP, CUBE	Advanced-Intermediate

Learning Path



Prerequisites

- Completed `02_SQL_Basics/` (SELECT, WHERE, ORDER BY, basic functions)
 - Comfortable with table relationships and foreign keys
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What You Will Be Able to Do

After completing this module:

1. Write complex multi-table JOIN queries confidently
 2. Aggregate data at any granularity using GROUP BY and HAVING
 3. Use subqueries and CTEs to break complex problems into steps
 4. Apply EXISTS/NOT EXISTS for efficient semi-join and anti-join patterns
 5. Combine result sets with UNION, INTERSECT, and EXCEPT
 6. Implement conditional logic with CASE expressions
 7. Build executive-level reports with ROLLUP and CUBE
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Key SQL Patterns Covered

Anti-Join (find rows with no match)

```
SELECT * FROM A WHERE NOT EXISTS (SELECT 1 FROM B WHERE B.id = A.id);
```

Semi-Join (find rows that have a match)

```
SELECT * FROM A WHERE EXISTS (SELECT 1 FROM B WHERE B.id = A.id);
```

Conditional Aggregation (pivot)

```
SELECT
  region,
  SUM(amount) FILTER (WHERE product = 'Widget A') AS widget_a,
  SUM(amount) FILTER (WHERE product = 'Widget B') AS widget_b
FROM sales GROUP BY region;
```

Running Total with ROLLUP

```
SELECT region, product, SUM(amount)
FROM sales
GROUP BY ROLLUP(region, product);
```

Interview Topics Covered

- All JOIN types and their behavior with NULLs
- NULL trap in NOT IN vs NOT EXISTS

- Difference between WHERE and HAVING
 - Correlated subquery performance (N+1 problem)
 - UNION vs UNION ALL performance
 - ROLLUP vs CUBE vs GROUPING SETS
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Next Module

Proceed to `04_Advanced_SQL/` for:

- Window functions (ROW_NUMBER, RANK, LAG, LEAD)
- CTEs and recursive queries
- LATERAL joins
- Advanced aggregation